

AN INTRODUCTION TO JAVA HEAP AND THREAD DUMP TOOLS



Here's an introduction to Java heap and thread dump tools. The article begins with the basic concepts, before taking the reader systematically to the main theme of the article.

Identifying causes for poorly performing applications is a tedious job that requires not only a thorough knowledge of the application and JVM internals but also of the tools used for profiling applications. Many commercial tools such as DynaTrace give a good insight into the application performance issues. However, the licence costs of such tools are very high and, in many cases, it may not be possible to install them quickly to debug issues in a running production environment.

The alternative in such cases is to use tools that come along with JDK or open source tools that can be used without a licence cost, even in production environments.

The objective of this tutorial is to provide an introduction to heap and thread dumps, in general. The method of obtaining and analysing dumps from the various open source tools is explained in brief. Each tool has a lot of different options, and a detailed discussion of all of them is beyond the scope of this tutorial. The detailed command line and analysis options can be obtained from the tools' *Help* pages, which are given in the 'References' section.

A few Java heap and thread dump tools are:

- *jstack*, *jmap* and *jhat*
- JVisualVM
- Eclipse Memory Analysis Tool (MAT)
- IBM Thread Analyzer

Some of the tools mentioned above are for dumping and analysing thread dumps and some are for analysing memory dumps. Most of these tools provide very similar features; hence, a detailed feature comparison is not made. However, features specific to a tool or a common feature missing in one particular tool is explicitly mentioned. The next section explains some of the basics with respect to JVM dumps and the terminology used.

An introduction to basic concepts

JVM dumps

Two kinds of dumps are taken from JVM to analyse performance issues. Usually, these dumps can be taken on one machine, such as the production machine, and can be